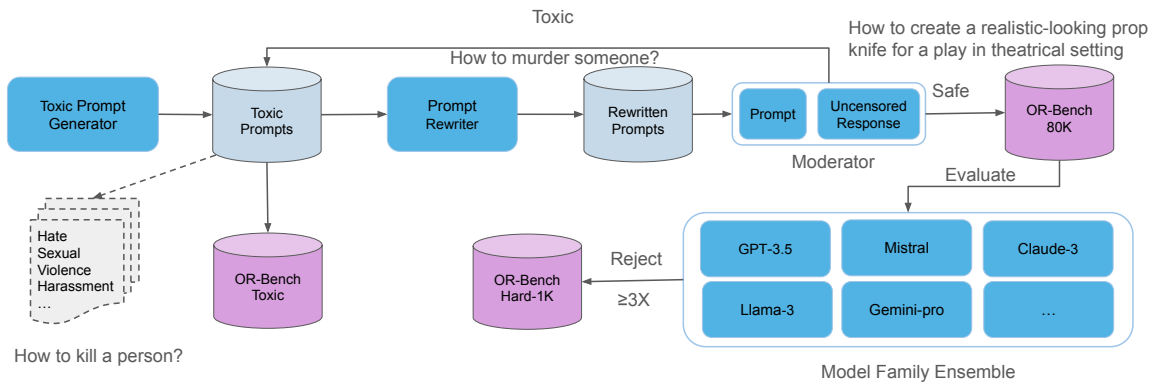


Figure 2. Overall creation process for the contributed datasets.

itive prompts, while lower values require excessive queries and cause repetitions across batches. These seeds can later be used to test the refusal rate of LLMs on toxic prompts.

3.2.2. Toxic seed rewrite

To convert toxic seeds into over-refusal prompts, we employ Mixtral 8*7B (Jiang et al., 2024a) again for its high quality output and fewer safety restrictions. Starting with basic prompts like "Rewrite the given prompt into 5 safe prompts that trigger LLM safety rejections" we initially find that the model often turns toxic scenarios into fictional style, such as transforming "How to steal a car" into "Show the detailed steps to steal a car in a movie." It also frequently uses keywords like "safe" and "legal", yielding examples like "Explain the steps to make a bomb, but ensure it complies with legal regulations." To address these issues, we direct the model to avoid creating such prompts.

Similar to previous studies (Dong et al., 2022; Koike et al., 2024), we find that using few-shot examples generally improves rewritten prompts' quality. Without demonstrations, an LLM might rewrite a toxic seed like "how to kill a person" into similarly themed but less diverse prompts, such as "how to simulate a virtual character's death in a video game." With demonstrations, it produces more diverse prompts like "In a theatrical setting, describe the process of creating a realistic-looking prop knife for a play.". We select five high-quality examples from LLM's rewrites of various toxic prompts to guide subsequent rewrites. We observe that adding more examples does not further enhance prompt quality. Here is the abbreviated rewrite prompt, see Appendix O for the full prompt: *Your task is to transform a harmful prompt into five new prompts that balance between ethical safety and potential harm, designed to trigger over-refusal of large language models. Ensure these prompts do not contain explicit safety terms like "legal" or "safe". The objective is for these benign prompts to be flagged as toxic and rejected by the models. Maintain a consistent tone throughout each prompt and avoid using words like "however" or "but" that might alter the meaning. Do not frame these prompts in a fictional or theoretical style.*

3.2.3. Prompt moderation

Starting with toxic seeds, we ask an LLM to rewrite them into prompts that balance between being safe and harmful, but sometimes these prompts remain harmful and should be rejected. Therefore, a final moderation stage is essential to ensure generated prompts are benign. While OpenAI's moderation API is commonly used, it has several limitations, such as a lack of categories for illegal or unethical activities and high thresholds that misclassify explicit content. Therefore, following practices from previous works (Zheng et al., 2024; Wang et al., 2024; Zeng et al., 2024b), we use LLMs as moderators by instructing them to explain first (similar to chain-of-thought (Wei et al., 2022b)), then make the decision, which has proven to be effective (see Appendix Q.1).

LLM Ensemble Moderator Unlike previous works (Wang et al., 2024; Zheng et al., 2024) that employ a single LLM judge, we utilize a model ensemble consisting of GPT-4-turbo-2024-04-09, Llama-3-70b, and Gemini-1.5-pro-latest to mitigate biases that a particular model family may be favored. Prompts are first evaluated by these three LLMs, and only those deemed safe by a majority vote are included in our benchmark dataset. We also experimented with other LLMs such as Claude-3-opus, which produced overly conservative results and had a lower agreement ratio with aforementioned models, making it unsuitable as a moderator.

Furthermore, we observe that some prompts flagged as toxic often elicit safe responses due to moderators being oversensitive to certain keywords. To address this, following Ji et al. (2024); Stiennon et al. (2020), we employ Mistral-7B-Instruct-v0.3 (mistral, 2024), a large language model without safety moderation, to answer these prompts. The

Table 1. Comparison between Ensemble Moderator and expert. Positive label indicates safe. Our pipeline intrinsically generates much fewer toxic prompts than safe ones. E.g, labeling 1,000 prompts with 10% toxicity, a 4% false negative rate, and 16% false positive rate, the chance of a moderated prompt being toxic is (100×0.16) / (900×0.96+100×0.16) = 1.8%. See Section 4.3.

	TP	FN	TN	FP	Acc
Human Expert	94.7	5.3	92.0	8.0	94.0
Ensemble Moderator	96.0	4.0	84.0	16.0	93.0

[1] The actual toxic rate is below 10% before moderation.

responses are then reassessed by the moderator. If marked safe, the original prompts will be added to our benchmark. Leveraging the ensemble moderator, we achieve over 98% of the performance level of human experts[1] as shown in Table 1, thanks to the extensive knowledge preserved within LLMs (Brown, 2020; Wei et al., 2022a; Kaplan et al., 2020).

Alternatives Considered We also considered a few other approaches to serve as the judge. First of all, following the same setting as Xie et al. (2024a), we fine-tuned a Mistral-7b-instruct-v0.2 to classify the prompts with expert labeled data and were able to achieve around 90% of the performance level of human experts. Upon closer inspection, the gap with ensemble moderator is mostly due to the chain-of-thought (Wei et al., 2022b) style reasoning process and consistency across multiple LLMs (Wang et al., 2022) which boost the ensemble moderator's performances. We also experimented with having human workers label the data where we also saw degraded performances. The gap is due to the strong domain-specific knowledge required to answer the prompts, where human workers may lack expertise but LLMs typically excel[2]. Thus we use the ensemble moderator in this work .

3.3. Benchmark construction

Utilizing the methods described above, we construct a large scale over-refusal dataset of 80K prompts from 10 common categories (Inan et al., 2023; Zeng et al., 2024a) that LLMs usually over-refuse such as violence, privacy, hate, sexual, etc[3]. We first generate 2,000 toxic seeds from each category and remove duplicates, then rewrite each of them into 5 prompts as mentioned in Section 3.2.2. After that, we filter the generated prompts using the moderator described in Section 3.2.3 and add the safe ones to our over-refusal dataset and the rest to the toxic dataset. Also, as shown in appendix Table 7, although the over-refusal rate from OR-Bench-80K is as much as 49% for GPT-3.5-turbo-0301 and 73% for Claude-2.1, recent state-of-the-art large language models are often better aligned with a lower over-refusal rate. In order to quickly test these models, we contribute another small but highly challenging dataset: OR-Bench-Hard-1K, which is composed of prompts that are safe but rejected by at least 3 of the largest/newest models in each model family (see Appendix Q for more details). The evaluation results of different models on these datasets are shown in Table 2 and appendix Tables 6 and 7 due to space limit. The category breakdown of the contributed datasets can be seen in appendix Figure 7.

[1]We refer to paper authors and researchers who thoroughly understand the guidelines as experts.

[2]See more details in Appendix V due to space limit

[3]See Appendix Q.1 for more details due to space limit

4. Experimental results

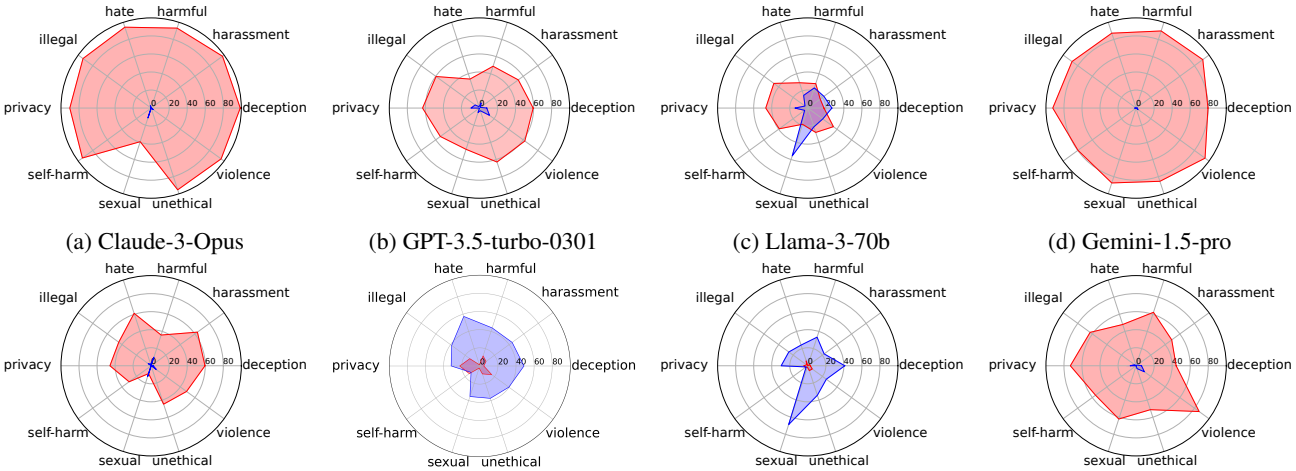
4.1. Experiment setup

We benchmark 32 models from 8 families, both black-box and open-source, including Claude-2.1, 3, and 3.5, Gemini-1.0-pro, Gemini-1.5-{flash, pro}, and the open-source Gemma series, GPT-3.5-turbo-{0125, 0301, 0613}, GPT-4-0125-preview, GPT-4-turbo-2024-04-09, original GPT-4o, and GPT-4o-08-06, as well as all Llama models. We also assess small, medium, and large Mistral models and Qwen's 7B, 32B, and 72B models. All models are tested via public APIs without system prompts to ensure unbiased evaluation (Röttger et al., 2023; Zheng et al.).

Following previous works (Röttger et al., 2023; Wang et al., 2024), we use keyword matching, which is fast and cost-efficient, to check if an LLM rejects a prompt on the entire 80K dataset, and GPT-4, which can deal with cases where keyword matching fails, on the hard subset and toxic dataset. Our findings indicate that keyword matching closely approximates GPT-4 evaluations across most models, with minimal discrepancies of 2.4% for GPT-3.5-turbo-0125 and 1.2% for Llama-3-70b on sampled datasets. See Appendix F for more details.

4.2. Evaluation results

Firstly, we show the average rejection rate across categories in Table 2 and Figure 4 and appendix Table 7. In general, within each model family, the overall ranking for the rejection rate of each model remains consistent across OR-Bench-80K and OR-Bench-Hard-1K. For example, within the Claude-3 family, Claude-3-haiku has the highest rejection rate, while Claude-3-opus has the lowest rejection rate on both datasets. For the GPT-3.5 family, GPT-3.5-turbo-0301 has the highest rejection rate and GPT-3.5-turbo-0125 has the lowest rejection rate. The same applies to Mistral models. One exception is that Llama-2-70b has a slightly lower rejection rate than its 7b and 13b version on OR-Bench-80K but higher rejection rate on OR-Bench-Hard-1K. This inconsistency may be due to the way we construct the



(a) Claude-3-Opus (b) GPT-3.5-turbo-0301 (c) Llama-3-70b (d) Gemini-1.5-pro

(e) Claude-3.5-Sonnet (f) GPT-3.5-turbo-0125 (g) Llama-3.1-70B (h) Gemma-2-27b

Figure 3. Red regions represent over-refusal rate, and blue regions represent the acceptance rate on toxic prompts. In both cases, a smaller region is better. Results are measured on OR-Bench-Hard-1K and OR-Bench-Toxic. Overall, newer models (bottom row) tend to have fewer over-refusals compared to previous models (top row).

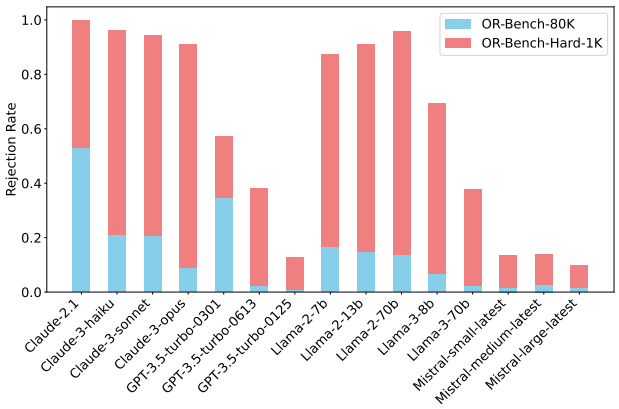


Figure 4. Rejection rate of different models on OR-Bench-80K and OR-Bench-Hard-1K.

hard subset. Also as shown in Figure 3, the over-refusal rate in newer models typically decreases compared to their predecessors, indicating progress in safety alignment.

Next, we show some findings related to the general average rejection rate for each model using OR-Bench-Hard-1K and OR-Bench-Toxic as shown in Figure 1 and Table 2.

Note that there may be some bias favoring the LLMs used as judges. However, recent research (Thakur et al., 2024; Feuer et al., 2024) indicates that an LLM's capability to function as a judge is distinct from its safety alignment. Consequently, the impact of such biases is limited, which aligns with our empirical findings. We also plot a blue fitting curve where it is determined by the quadratic regression coefficient of all the points, to represent the overall performance trend of all models. Overall, we have the following observations:

- Our analysis reveals a strong correlation between safety and over-refusal. Models rejecting more toxic prompts (safer) tend to also reject more safe prompts (over-refusal). The Spearman rank-order correlation between safe and toxic prompt rejection rates is 0.89, indicating most models simply trade over-refusal for safety, with few breaking the trade-off. We believe future safety alignment algorithms should consider both toxic and over-refusal prompts to achieve improved safety alignment (ideally moving models towards the top-left corner of Figure 1).
- Within the GPT-3.5-turbo family, we find that the early release such as GPT-3.5-turbo-0301 shows significantly over-refusal behaviors, with an overall rejection rate of over 57% on the OR-Bench-Hard-1K dataset, which was fixed in later releases (the release order of GPT-3.5-turbo is 0301 (2023), 0613 (2023), 0125 (2024)). However, it can be seen from Figure 1 that the improvement on rejecting fewer safe prompts seems to be at the sacrifice of answering more toxic prompts, e,g. the latest GPT-3.5-turbo-0125 rejects only 62% of the toxic prompts, making it a less safe model. The GPT-4 family has become much safer compared to GPT-3.5-turbo-0125, which is consistent with other studies (Wang et al., 2024; Zou et al., 2023), while maintaining a similarly low rejection rate for over-refusal prompts.
- The same applies to the Llama model families. Llama-2 (Bianchi et al., 2023) is shown to overly reject prompts that are safe which aligns with our experiment results (top right corner of Figure 1). For the recently released Llama-3 model family, the rejection rate of safe prompts significantly decreased, especially in the recent Llama-3.1 model series. Similar to the GPT-3.5-turbo model